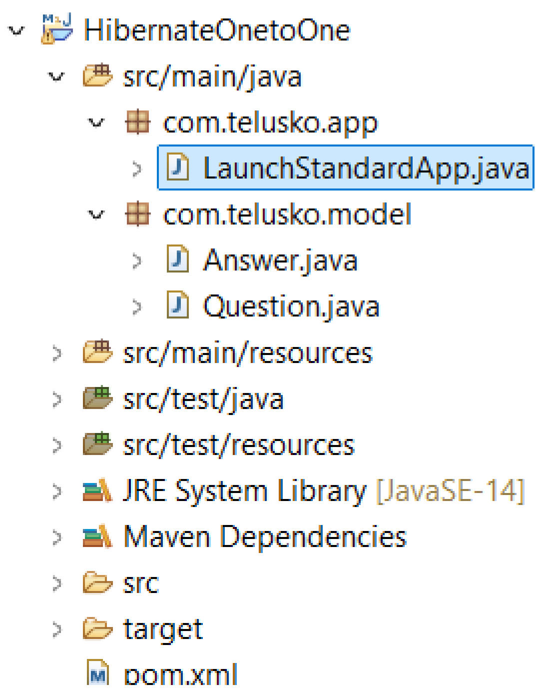
Hibernate One-to-One Association

- **Definition**: A one-to-one association refers to a relationship where one entity instance is associated with exactly one instance of another entity.
- Project Structure



Example in Code:

Student.java

```
package com.telusko.model;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.OneToOne;

@Entity
```

```
public class Question
{
        @Id
        @Column(name="question_id")
    private Integer id;

    private String question;

    @OneToOne(cascade=CascadeType.ALL)
    private Answer answer;

    public Question()
    {
         System.out.println("Zero Param Constructor of Question");
    }

        @Override
        public String toString() {
                return "Question [id=" + id + ", question=" + question + ", answer="
+ answer + "]";
        }

        public Integer getId() {
                return id;
        }

        public void setId(Integer id) {
                this.id = id;
        }

        public String getQuestion() {
                return question;
        }
```

```
	public void setQuestion(String question) {
		this.question = question;
	}

	public Answer getAnswer() {
		return answer;
	}

	public void setAnswer(Answer answer) {
		this.answer = answer;
	}


}
```

Answer.java

```
package com.telusko.model;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.OneToOne;

@Entity
public class Answer
{
	@Id
	@Column(name="answer_id")
    private Integer id;
    private String answer;
```

```
    public Question getQuestion() {
                return question;
        }
        public void setQuestion(Question question) {
                this.question = question;
        }

        @OneToOne(cascade=CascadeType.ALL)
    private Question question;

    public Answer()
    {
         System.out.println("Zero param Constructor of Answer");
    }
        public Integer getId() {
                return id;
        }

        public void setId(Integer id) {
                this.id = id;
        }

        public String getAnswer() {
                return answer;
        }

        public void setAnswer(String answer) {
                this.answer = answer;
        }
        @Override
        public String toString() {
                return "Answer [id=" + id + ", answer=" + answer + ", question=" +
question + "]";
```

```
        }


}
```

```
package com.telusko.app;

import org.hibernate.HibernateException;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.cfg.Configuration;

import com.telusko.model.Answer;
import com.telusko.model.Question;

public class LaunchStandardApp
{

        public static void main(String[] args)
        {
                Configuration config = null;
                SessionFactory sessionFactory = null;
                Session session = null;
                Transaction transaction = null;
                boolean flag=false;

                config=new Configuration();

                config.configure();

                sessionFactory=config.buildSessionFactory();
```

```
		session=sessionFactory.openSession();

		Question q1=new Question();
		q1.setId(1);
		q1.setQuestion("What is Hibernate?");

		Answer answer1 =new Answer();
		answer1.setId(1);
		answer1.setAnswer("Hibernate is an ORM Framewok");

		q1.setAnswer(answer1);
		answer1.setQuestion(q1);



		Question q2=new Question();
		q2.setId(2);
		q2.setQuestion("What is JPA?");

		Answer answer2 =new Answer();
		answer2.setId(2);
		answer2.setAnswer("It is a specificatiion used to persist "
				+ "data between Java object and relational database.");

		q2.setAnswer(answer2);
		answer2.setQuestion(q2);

//		Question qa=session.get(Question.class, 1);
//		System.out.println(qa);

		try
		{
			transaction=session.beginTransaction();
			//session.save(student);
```

```
                    session.persist(q1);
                    session.persist(q2);

//                  session.persist(answer1);
//                  session.persist(answer2);
                flag=true;

            }
            catch(HibernateException e)
            {
                    e.printStackTrace();
            }
            catch(Exception e)
            {
                    e.printStackTrace();
            }

            finally
            {
                    if(flag==true)
                    {
                            transaction.commit();
                    }
                    else
                    {
                            transaction.rollback();
                    }

                    session.close();
                    sessionFactory.close();

            }

        }
```

```
}
```

Output:

```
mysql> drop database telusko_db;
Query OK, 6 rows affected (0.10 sec)

mysql> create database telusko_db;
Query OK, 1 row affected (0.01 sec)

mysql> show tables;
ERROR 1046 (3D000): No database selected
mysql> use telusko_db;
Database changed
mysql> show tables;
+----------------------+
| Tables_in_telusko_db |
+----------------------+
| answer               |
| question             |
+----------------------+
2 rows in set (0.00 sec)
```

```
mysql> select * from question;
+------------------+-------------+--------------------+
| answer_answer_id | question_id | question           |
+------------------+-------------+--------------------+
|                1 |           1 | What is Hibernate? |
|                2 |           2 | What is JPA?       |
+------------------+-------------+--------------------+
2 rows in set (0.00 sec)

mysql> select * from answer;
+-----------+----------------------+-----------------------------------------------------------------------------------------+
| answer_id | question_question_id | answer
      |
+-----------+----------------------+-----------------------------------------------------------------------------------------+
|         1 |                    1 | Hibernate is an ORM Framewok
      |
|         2 |                    2 | It is a specificatiion used to persist data between Java object and relational database. |
+-----------+----------------------+-----------------------------------------------------------------------------------------+
2 rows in set (0.00 sec)
```

```
Hibernate:
    drop table if exists Answer
Hibernate:
    drop table if exists Question
Hibernate:
    create table Answer (
        answer_id integer not null,
        question_question_id integer,
        answer varchar(255),
        primary key (answer_id)
    ) engine=InnoDB
Sept 15, 2024 12:23:41 PM org.hibernate.resource.transaction.backend.jdbc.internal.D
INFO: HHH10001501: Connection obtained from JdbcConnectionAccess [org.hibernate.engi
Hibernate:
    create table Question (
        answer_answer_id integer,
        question_id integer not null,
        question varchar(255),
        primary key (question_id)
    ) engine=InnoDB
Hibernate:
    alter table Answer
       add constraint UK_ljqm891ywk3128u2dyfkmc69d unique (question_question_id)
Hibernate:
    alter table Question
       add constraint UK_9y1pwogen3tjh3k2r2spard44 unique (answer_answer_id)
```

```
Hibernate:
    alter table Answer
       add constraint FK8yiifgngf37mxn437tqdab3rg
       foreign key (question_question_id)
       references Question (question_id)
Hibernate:
    alter table Question
       add constraint FKs6ghcwuovtcp489oo5dy7rvk5
       foreign key (answer_answer_id)
       references Answer (answer_id)
Zero Param Constructor of Question
Zero param Constructor of Answer
Zero Param Constructor of Question
Zero param Constructor of Answer
```

```
Hibernate:
    insert
    into
        Answer
        (answer, question_question_id, answer_id)
    values
        (?, ?, ?)
Hibernate:
    insert
    into
        Question
        (answer_answer_id, question, question_id)
    values
        (?, ?, ?)
Hibernate:
    insert
    into
        Answer
        (answer, question_question_id, answer_id)
    values
        (?, ?, ?)
Hibernate:
    insert
    into
        Question
        (answer_answer_id, question, question_id)
    values
        (?, ?, ?)
```

```
Hibernate:
    update
        Answer
    set
        answer=?,
        question_question_id=?
    where
        answer_id=?
Hibernate:
    update
        Answer
    set
        answer=?,
        question_question_id=?
    where
        answer_id=?
```

Explanation:

- The @OneToOne annotation specifies a one-to-one relationship between Question and Answer.

- CascadeType.ALL: Operations like save or delete performed on the Question entity are cascaded to the Answer entity.
- If CascadeType.ALL is not used, you need to save the Answer object separately.
- This relationship can work as either unidirectional or bidirectional.